Supporting Information

Transfer contrastive learning for Raman spectra data of urine: Detection of glucose, protein, and prediction of kidney disorders

Zhuangwei Shi[a, #], Jiale Wang[b, #], Yunhao Su[a], Xiaohong Liang[a], Jianchen Zi[a,*], Chenhui Wang[a,c,*], Hai Bi[a,*], Xia Xiang[b,*]

[a] XLab, Ji Hua Laboratory, Foshan, Guangdong, 528200, China.

[b] Department of Nursing, The First People's Hospital of Foshan, Foshan, Guangdong, 528000, China.

[c] Personalized Nutrition, Baden-Württemberg Cooperative State University Heilbronn, Heilbronn, Baden-Württemberg, 74076, Germany.

[#]Equal contributions.

*Corresponding authors: zijianchen@jihualab.com (J. Zi); wangchh@jihualab.com (C. Wang); bihai@jihualab.com (H. Bi); xiangxiafoshan@163.com (X. Xiang).

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Explanation of contrastive loss

Following previous research [1], in our work, z'_{2i-1} and z'_{2i} are representations obtained from different augmentation (in this paper we adopt random mask for Raman spectrum) of the same spectrum x_i. Note that the contrastive loss

$$\ell(p,q) = -\log \frac{\exp(s(z'_p, z'_q)/\tau)}{\sum_{r=1}^{2n} \beta_{pr} \exp(s(z'_p, z'_r)/\tau)}$$

Here, $s(z'_p, z'_q)$ denotes the cosine similarity between z'_p and z'_q, τ is hyperparameter

$$s(z'_p, z'_q) = \frac{z'^T_p z'_q}{\left\|z'_p\right\| \cdot \left\|z'_q\right\|}$$

$$\beta_{pr} = \begin{cases} 1 & p \neq r \\ 0 & p = r \end{cases}$$

Apparently, $s(z'_p, z'_q)$ is ranging in [-1,1]. When z'_p and z'_q are more similar, $s(z'_p, z'_q)$ is larger and closer to 1.

Let $p = 2i-1$ and $q = 2i$, when the similarity between z'_{2i-1} and z'_{2i} is maximized, $\exp(s(z'_p, z'_q)/\tau)$ will be larger. When the similarity between z'_{2i-1} and other representations z'_r $(r \neq p)$ is minimized, $\sum_{r=1}^{2n} \beta_{pr} \exp(s(z'_p, z'_r)/\tau)$ will be smaller. Hence, $\ell(2i-1, 2i)$ will be smaller. Likewise, let $p = 2i$ and $q = 2i-1$, when the similarity between z'_{2i-1} and z'_{2i} is maximized, and the similarity between z'_{2i} and other representations z'_r $(r \neq p)$ is minimized, $\ell(2i, 2i-1)$ will be smaller.

Therefore, for each sample $i = 1,2,\dots,n$, contrastive representation learning aims to maximizing the similarity between z'_{2i-1} and z'_{2i}, while minimizing the similarity between z'_{2i-1} (or z'_{2i}) and all other samples by minimizing

$$L_c = \frac{1}{2n} \sum_{i=1}^{n} \left(\ell(2i-1, 2i) + \ell(2i, 2i-1)\right)$$

Explanation of focal loss

Let $\hat{Y}$ and Y denote the true label matrix and predicted label matrix, respectively. Here, $\widehat{y_i}$ is the i-th row of $\hat{Y}$, denoting the one-hot coding of the i-th sample's true label, i.e.,

$$\widehat{y_{ij}} = \begin{cases} 1 & sample\ i\ belongs\ to\ class\ j \\ 0 & else \end{cases}$$

Similarly, y_i denotes the predicted label of the i-th sample. The activation function of the last layer of classifier module is set as softmax function, i.e., let u_i denotes the i-th row in the output of the last layer, then

$$y_{ij} = softmax(u_{ij}) = \frac{\exp u_{ij}}{\sum_k \exp u_{ik}}$$

Hence, y_{ij} denotes the probability of sample i belongs to class j, $y_{ij} > 0$ and $\sum_k y_{ik} = 1$. Apparently, if sample i belongs to class j, the probability can be computed by

$$p_i = y_i^T y_i$$

Following previous research [2], focal loss can be defined as

$$L_f = -\frac{1}{n}\sum_{i=1}^{n}(1-p_i)^{\gamma}\log p_i$$

Here, γ is hyperparameter, and n is the number of samples.

Explanation of contrastive learning with focal loss

TCRaman adopts total loss by combining contrastive loss and focal loss

$$L = L_c + \alpha L_f$$

Here α is a hyperparameter to balance focal loss and contrastive loss.

Cross entropy loss is the most commonly used classification loss. Using the definition of p_i, the cross entropy loss for binary classification can be written as

$$L_b = -\frac{1}{n}\sum_{i=1}^{n} \log p_i$$

Compared with cross entropy loss, focal loss adopts a weight $(1 - p_i)^\gamma$ to each sample, samples with lower p_i are with higher weights. Apparently, lower p_i denotes more challenging samples for classification. As the expectation of the initial p_i at the beginning of training is the proportion of the class which the sample belongs to in the dataset, samples in small class are with higher weights. Hence, the supervised contrastive learning with focal loss enhances the ability of representation learning for imbalanced classification.

Details of sampling methods

As sampling is a common way to tackle imbalanced dataset, we also compare TCRaman's performance using different sampling methods. Generally, there are two sampling strategies. Over-sampling generates new samples by sampling the minority samples with replacement, while under-sampling reduces the majority samples by selectively sampling. SMOTE (Synthetic Minority Oversampling Technique) is adopted for over-sampling, and Tomek link is adopted for under-sampling. We adopt a Python library, named imbalanced-learn (version 0.13.0), to implement the sampling methods.

In our study, negative samples are more than positive samples. Therefore, SMOTE (over-sampling) is designed to re-sample positive samples in training set, until the number of positive samples is equal to negative samples. For instance, in glucose detection task, the numbers of negative and positive samples in training set are 1080 and 228, respectively. Thus, SMOTE over-samples positive samples until the number reaches 1080.

In order to fully test the generalizability of the proposed method, and discuss the impact of the number of modeling samples, we design different under-sampling ways.

- Tomek1: Under-sample negative samples in training set, until the number of positive samples is equal to negative samples. For instance, in glucose detection task, the numbers of negative and positive samples in training set are 1080 and 228, respectively. Thus, Tomek1 under-samples negative samples until the number reaches 228.
- Tomek2: Under-sample both negative and positive samples in training set, until both the numbers of positive and negative samples are 100.
- Tomek3: Under-sample positive samples in training set, until the number of positive samples is 100. Note that Tomek3 provides a more challenging scenario to test the robustness of TCRaman.

The results are shown in Fig. S11. SMOTE, Tomek1 and Tomek2 are common ways to tackle imbalanced dataset by keeping the number of positive samples is equal to negative samples using sampling. TCRaman with none of these sampling methods performs best, demonstrating that TCRaman excels at extracting information from the original and unsampled dataset. In contrast, Tomek3 is a more challenging scenario. However, TCRaman can still keep quite high accuracy when using Tomek3, highlighting its robustness.

Tables

Table S1: Details of layers in TCRaman.

Module	Layer	Dimension of this layer's output
CNN	Convolutional layer (Conv1)	360
	ReLU activation function	360
	Max pooling layer (MaxPool1)	180
	Convolutional layer (Conv2)	180
	ReLU activation function	180
	Max pooling layer (MaxPool2)	90
MLP	Fully connected layer	32
	Output layer	c
	Softmax activation function	c

Note: As shown in Fig. 1(c), the input of Conv1 is the masked spectrum ${x_{2i-1}}'$ or x'_{2i}. Correspondingly, fully connected layer outputs representation ${z_{2i-1}}'$ or z'_{2i}. Then, final representation

$$z_i = \frac{z'_{2i-1} + z'_{2i}}{2}$$

Then, input z_i to output layer.

Besides, c denotes number of classes. In pretraining (bacterial Raman spectra dataset), $c = 30$. In finetuning (urine Raman spectra dataset), $c = 2$.

Table S2: Hyperparameters of the compared models. The other hyperparameters are kept as the default values in Scikit-learn, XGBoost, LightGBM or PyTorch libraries.

Model	Parameter	Candidate values	Best value
PCA-LDA	n_components	8, 16, 32, 64	32
PLS-DA	n_components	8, 16, 32, 64	32
SVM	gamma	0.001, 0.01, 0.1	0.01
	C	1, 2, 4	2
RF	max_depth	3, 5, 10, 20	5
	n_estimators	20, 50, 100, 150	100
XGBoost	max_depth	3, 5, 10, 20	5
	colsample_bytree	0.1, 0.3, 0.5, 0.7	0.5
	n_estimators	20, 50, 100, 150	100
LightGBM	max_depth	3, 5, 10, 20	5
	colsample_bytree	0.1, 0.3, 0.5, 0.7	0.3
	n_estimators	20, 50, 100, 150	100
CNN	num_layers	1, 2, 3	2
	kernel_size	3, 5, 7	5
LSTM	num_layers	1, 2, 3	2
	output_dim	45, 90, 180	90

Note: Hyperparameters tuning on CNN is performed on number of layers and size of convolutional kernel. Hyperparameters tuning on LSTM is performed on number of layers and output dimension. In CNN, the pooling size is commonly fixed to 2, as the input dimension of spectrum is fixed to 360, let ϕ denotes the number of layers, the output dimension is computed by $360/2^{\phi}$.

Table S3: To further demonstrate the validity of transfer learning and contrastive learning, we conduct transfer learning on basic CNN, LSTM and Transformer models using in the comparison with current methods. We also list the results of contrastive learning model without or with transfer learning (i.e., CRaman and TCRaman). Deep learning models with transfer learning can achieve higher accuracy, demonstrating the validity of transfer learning. However, TCRaman still performs better than the transfer learning version of CNN, LSTM and Transformer, demonstrating the validity of contrastive learning.

Model	Transfer learning	Glucose	Protein	Kidney disorders
CNN	No	0.818	0.873	0.882
	Yes	0.862	0.883	0.914
LSTM	No	0.802	0.856	0.873
	Yes	0.836	0.871	0.905
Transformer	No	0.870	0.882	0.895
	Yes	0.885	0.890	0.913
CRaman	No	0.890	0.886	0.905
	Yes	0.912	0.908	0.954

Table S4: Homogeneity of variance (HOV) test and t-test results between glucose, protein or double positive labels and kidney disorders labels in the whole dataset of urine spectra.

p-value	Glucose	Protein	Double
HOV test	0.218	0.242	0.276
t-test	0.066	0.184	0.002

Note: The HOV test of all three positive labels (glucose, protein and double positive) show that $p>0.05$, ensuring the homogeneity to conduct t-test. For t-test results, either glucose or protein positive is not significantly correspond to kidney disorders positive ($p>0.05$), while double-positive is significantly correspond to kidney disorders positive ($p=0.002$).

Table S5: The numbers of training and testing samples in different age groups. We have also conducted case studies by computing TCRaman's accuracy performance of different age groups.

Age	Number of training samples	Number of testing samples	Glucose	Protein	Kidney disorders
<40	165	120	0.908	0.908	0.958
40~49	138	129	0.915	0.907	0.953
50~59	318	219	0.909	0.904	0.954
60~69	354	255	0.910	0.914	0.953
70~79	288	156	0.917	0.910	0.955
≥80	45	48	0.917	0.896	0.958
All	1308	927	0.912	0.908	0.954

Figures

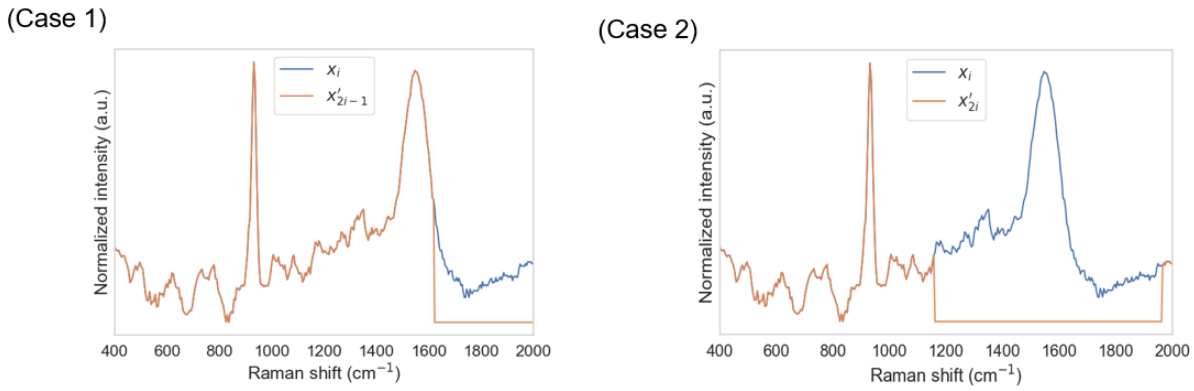


Figure S1: The example show the differences between the data processed with and without the mask.

Note: Let m denotes the dimension of spectrum (in this paper $m = 360$), a is an integer sampled randomly from a uniform distribution of $[0, m)$, and d is a number sampled randomly from a uniform distribution of [0.3,0.7]. Then, b is computed by Eq. (1) in this paper. Masking the $[a, b)$ interval of this spectrum is performed by setting the intensity values in the interval to 0.

Case 1 (mask x_i to get x'_{2i-1}): suppose $a = 250, d = 0.4$, so

$$a > (1 - d)m, b = m = 360$$

Mask the $[250,360)$ interval of x_i.

Case 2 (mask x_i to get x'_{2i}): suppose $a = 150, d = 0.5$, so

$$a \leq (1 - d)m, b = int(a + dm) = 330$$

Mask the $[150,330)$ interval of x_i.

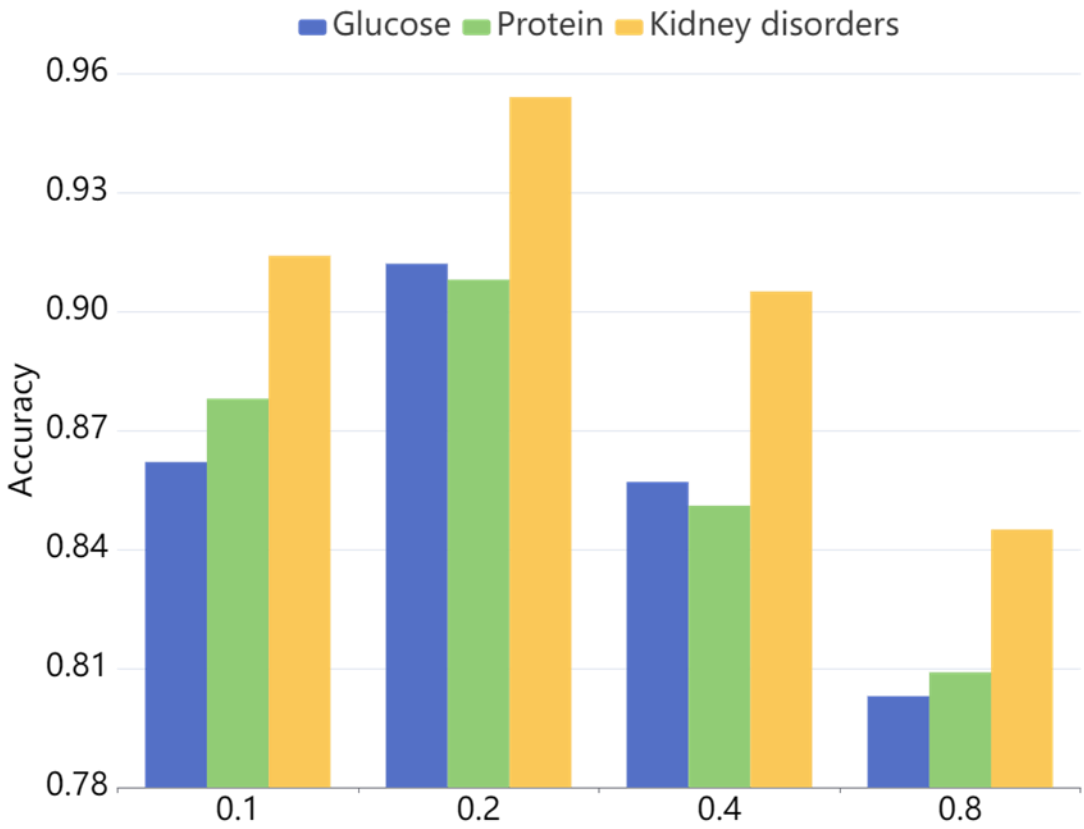


Figure S2: Accuracy results under different hyperparameter τ.

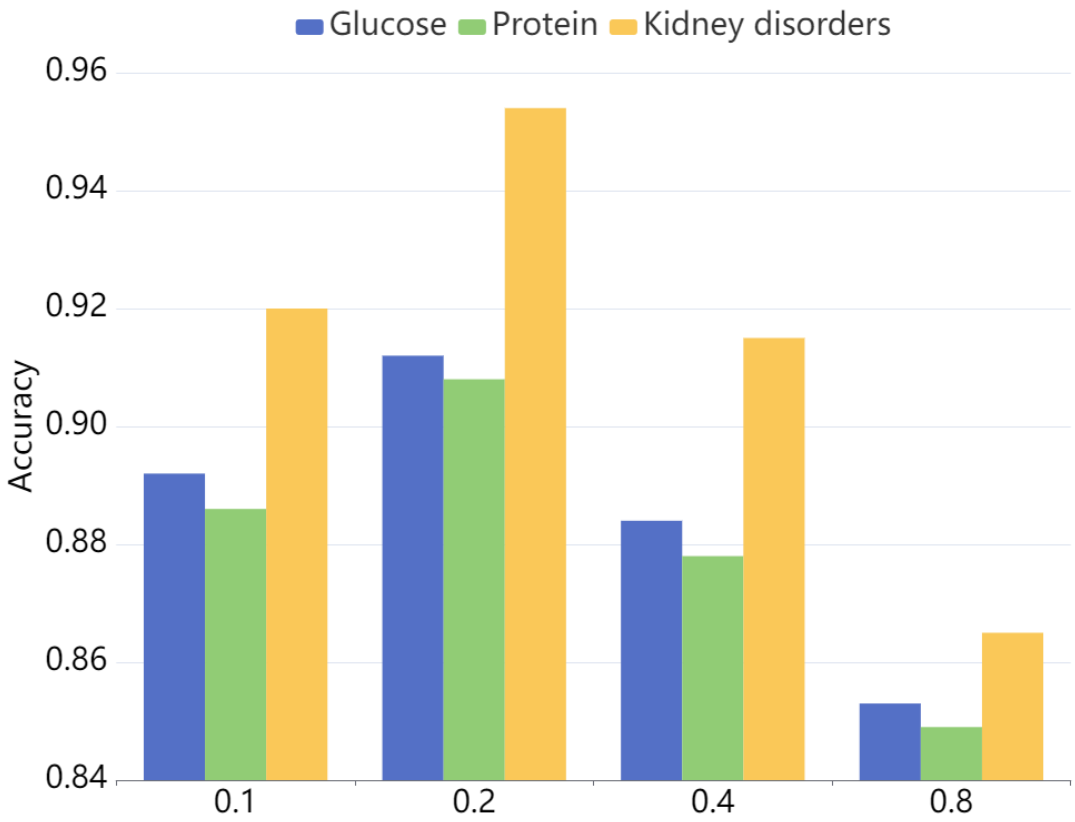


Figure S3: Accuracy results under different hyperparameter α.

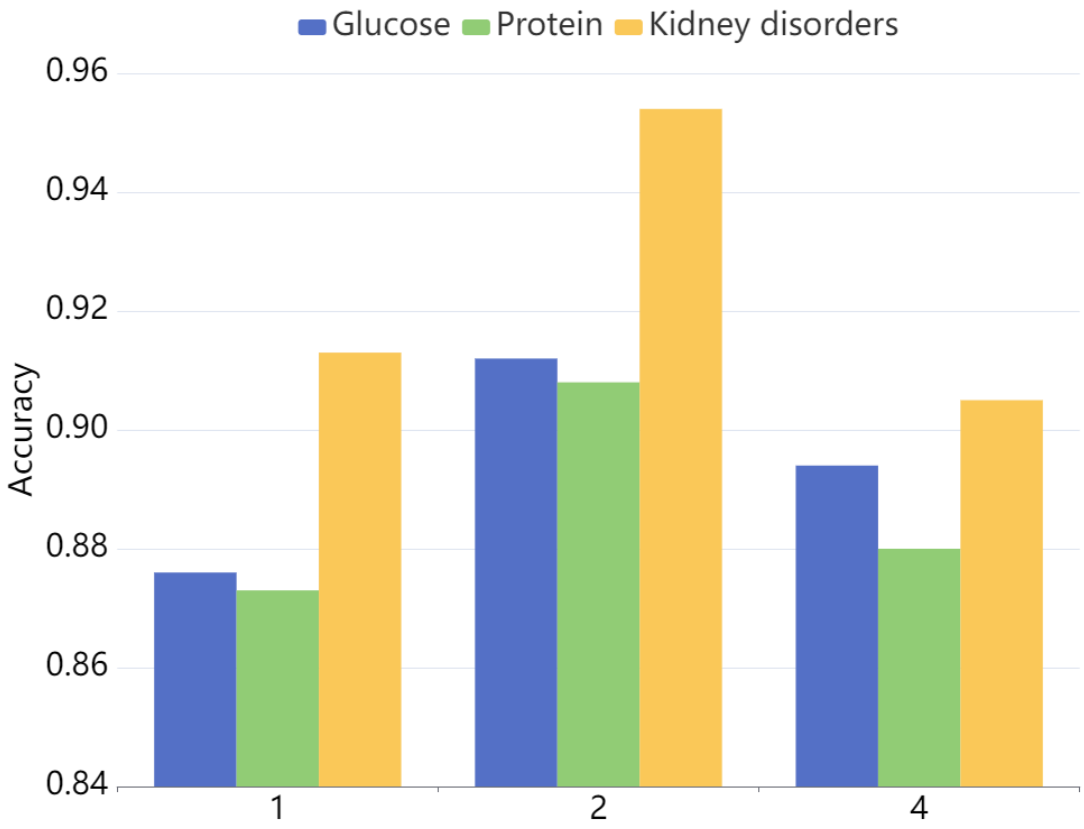


Figure S4: Accuracy results under different hyperparameter γ.

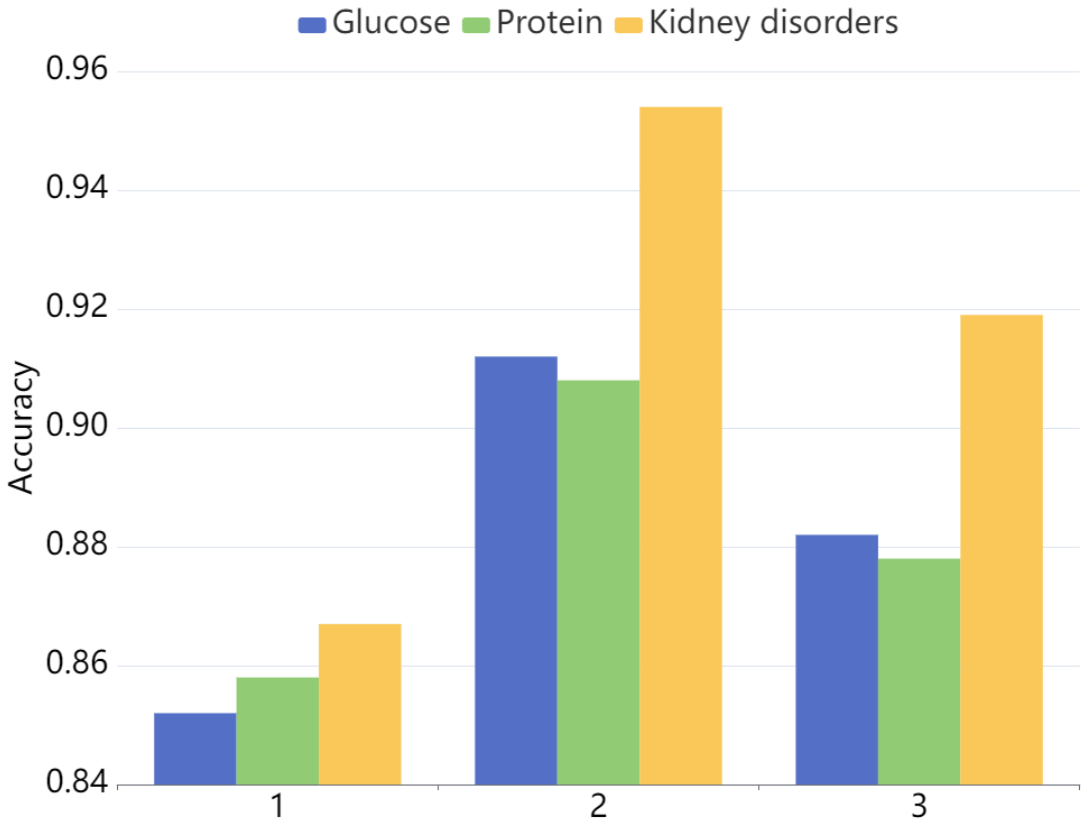


Figure S5: Accuracy results using different numbers of convolutional layers in representation module.

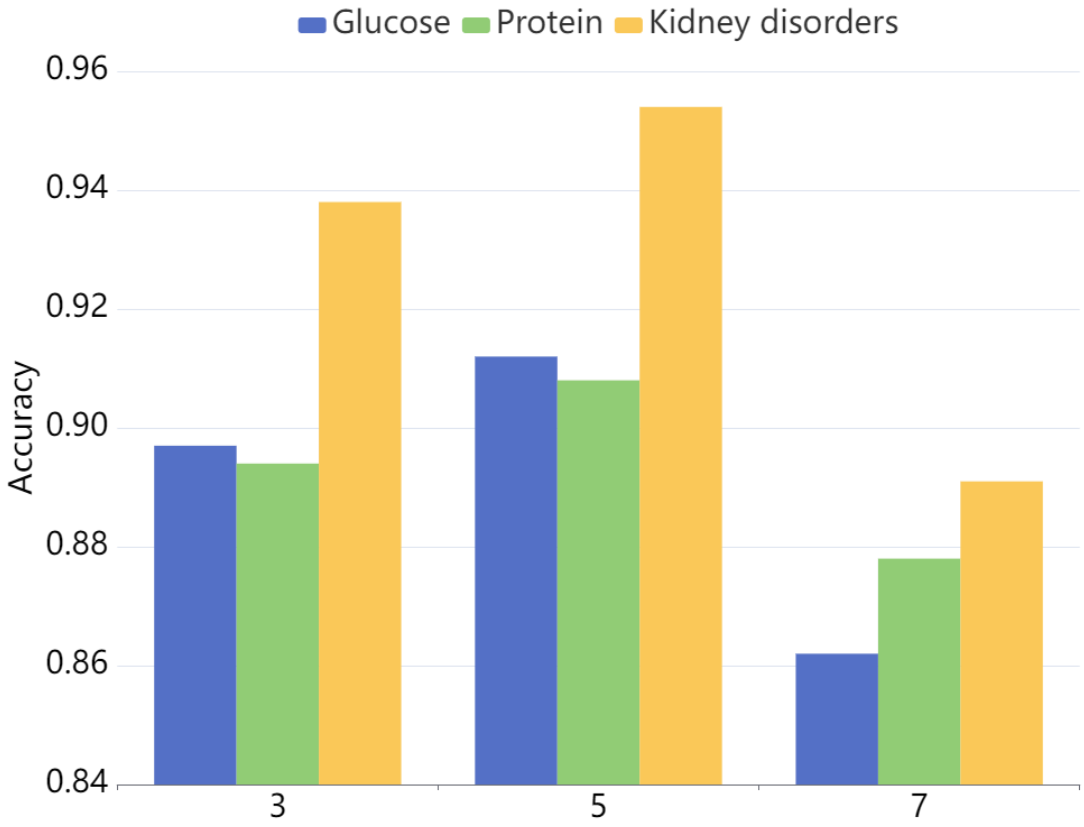


Figure S6: Accuracy results using different sizes of convolutional kernels in representation module.

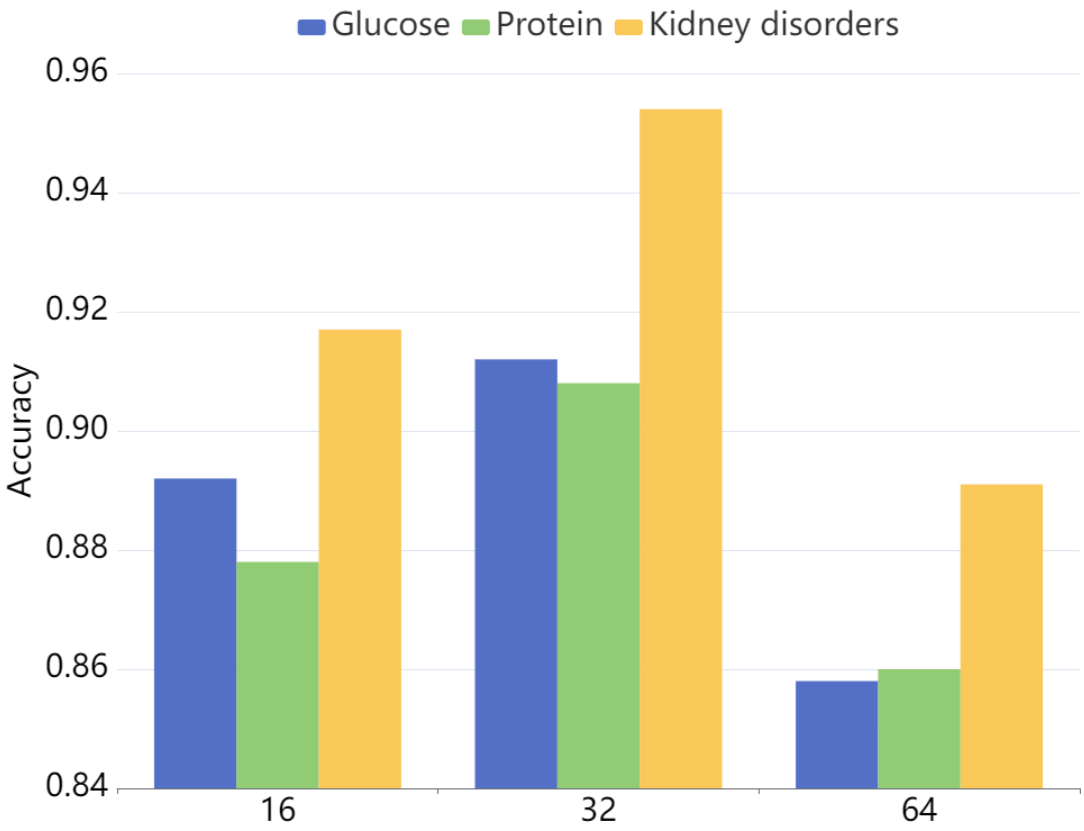


Figure S7: Accuracy results using different dimensions of representations.

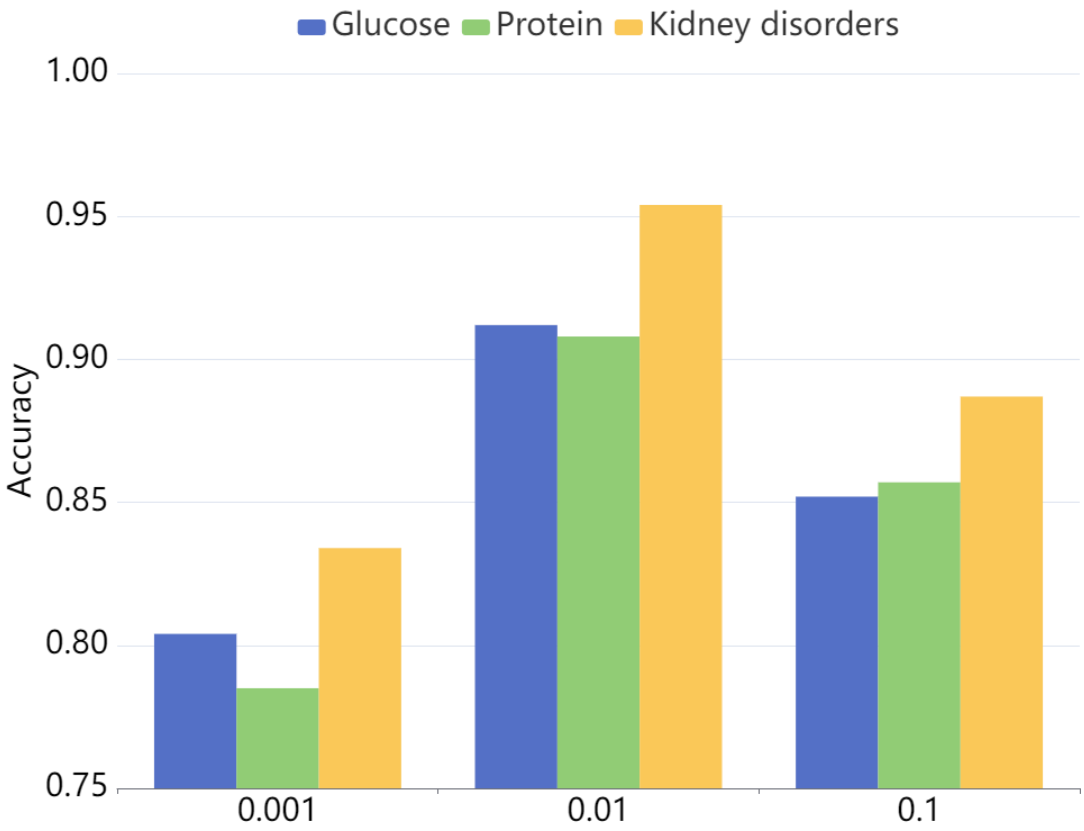


Figure S8: Accuracy results using different learning rates.

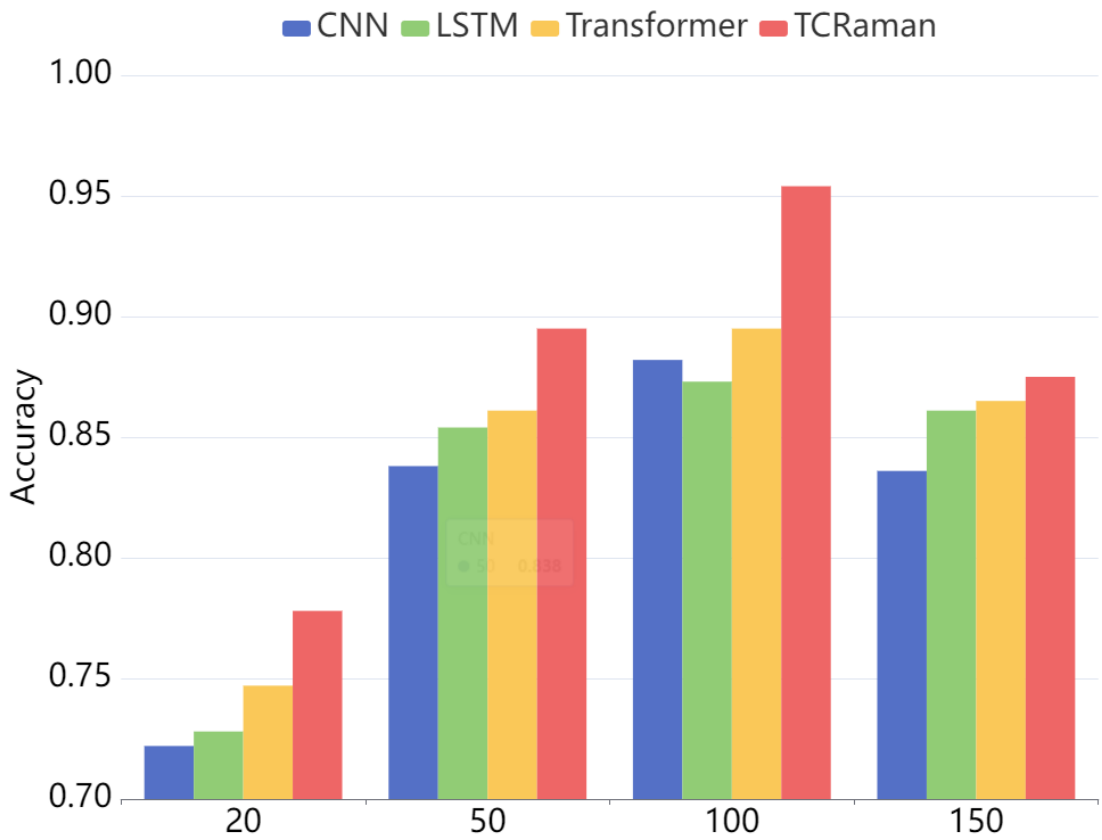


Figure S9: Accuracy results on kidney disorders detection using different epochs of CNN, LSTM, Transformer and TCRaman.

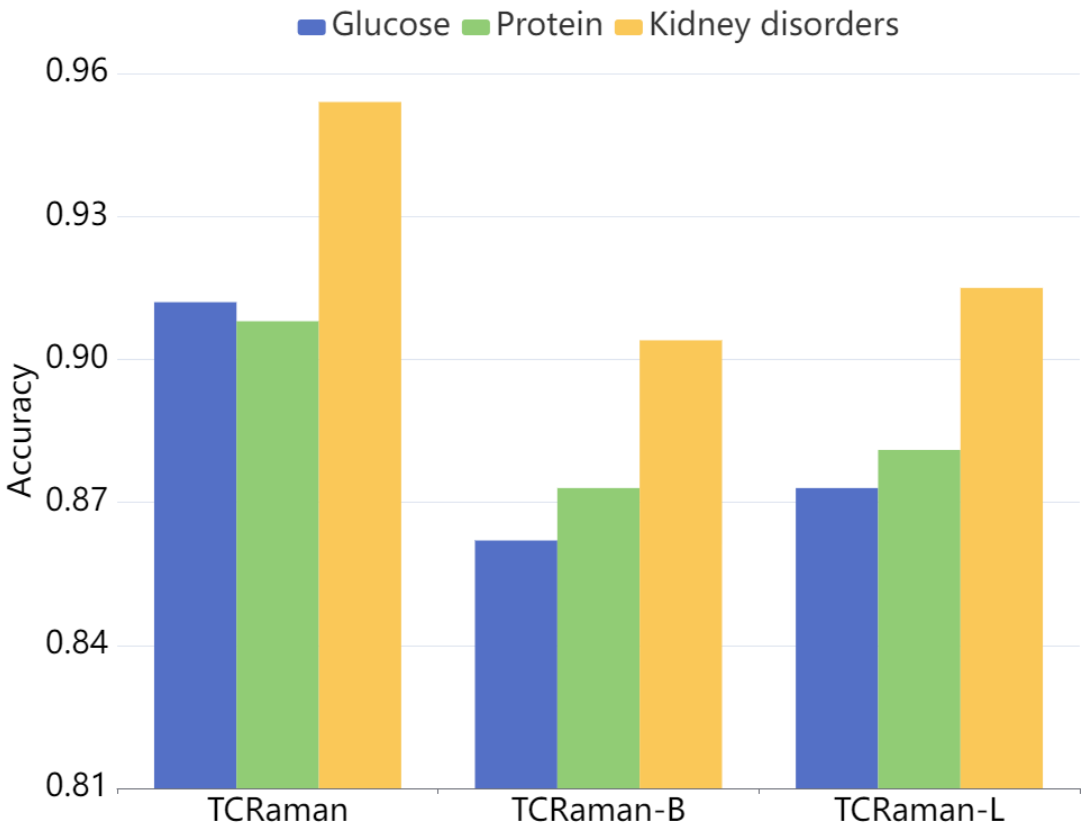


Figure S10: Ablation studies on contrastive learning model.

Note: In TCRaman-B, we replace the focal loss L_f with binary cross entropy loss L_b. In TCRaman-L, we replace the CNN module in TCRaman with LSTM. On all three detection tasks of glucose, protein and kidney disorders, TCRaman achieve highest accuracy. The results demonstrate that focal loss enhances the accuracy of contrastive learning for imbalanced classification tasks. On the other hand, previous studies in the field of computer vision [1,3] have shown that compared with LSTM, CNN can better perform the contrastive learning framework for representation learning. Herein, we verify this proposition on Raman spectral data.

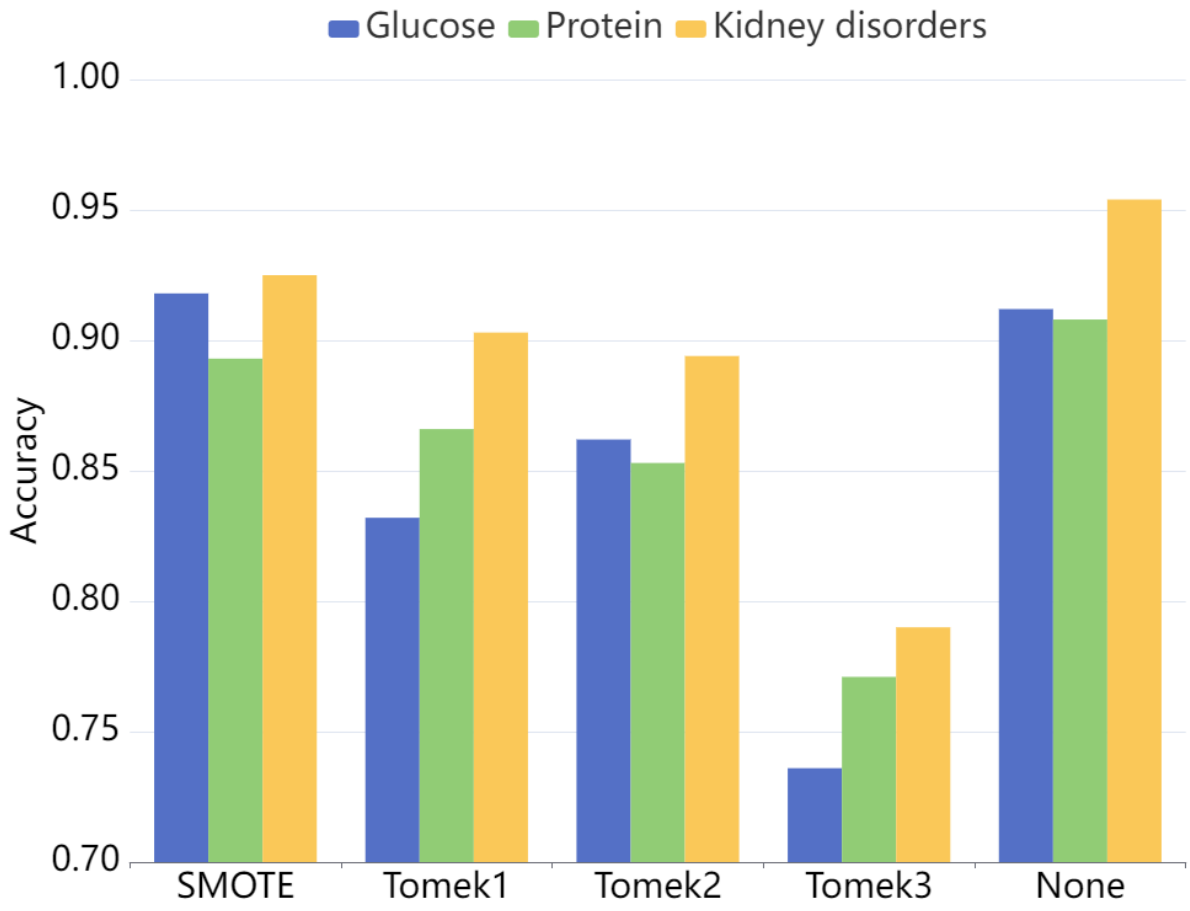


Figure S11: As sampling is a common way to tackle imbalanced dataset, we also compare TCRaman's accuracy performance using different sampling methods (see Details of sampling methods in Supporting Information).

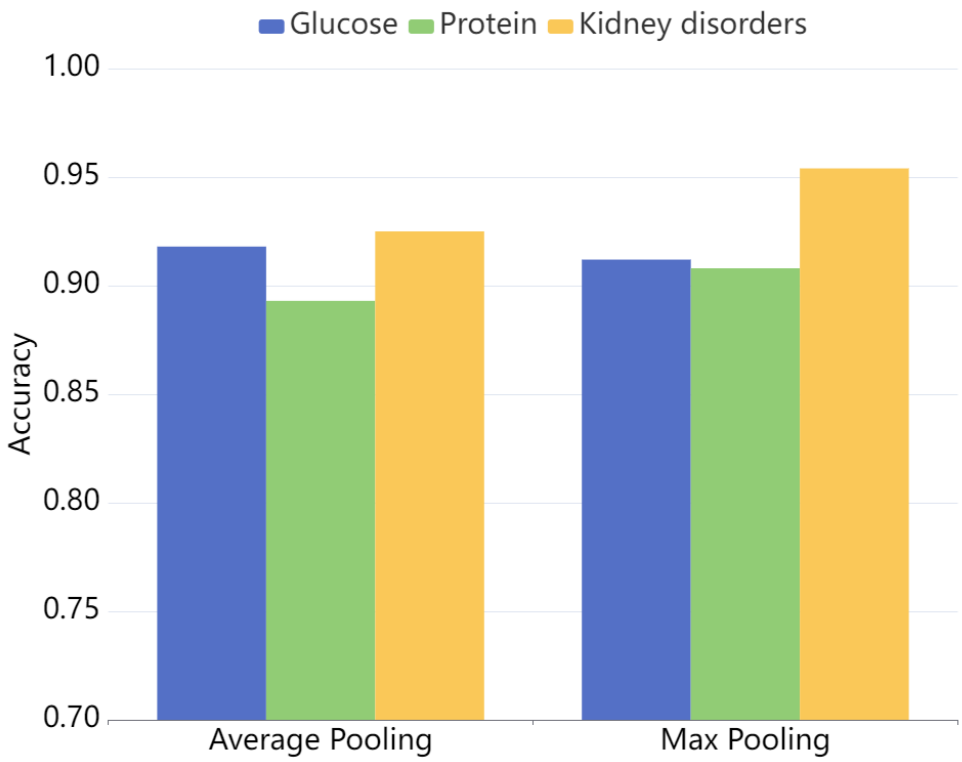


Figure S12: Except for max pooling, the average pooling is another widely-used pooling method for CNN. We have compared the performance of TCRaman model under different pooling methods, demonstrating that max pooling is more suitable in our study.

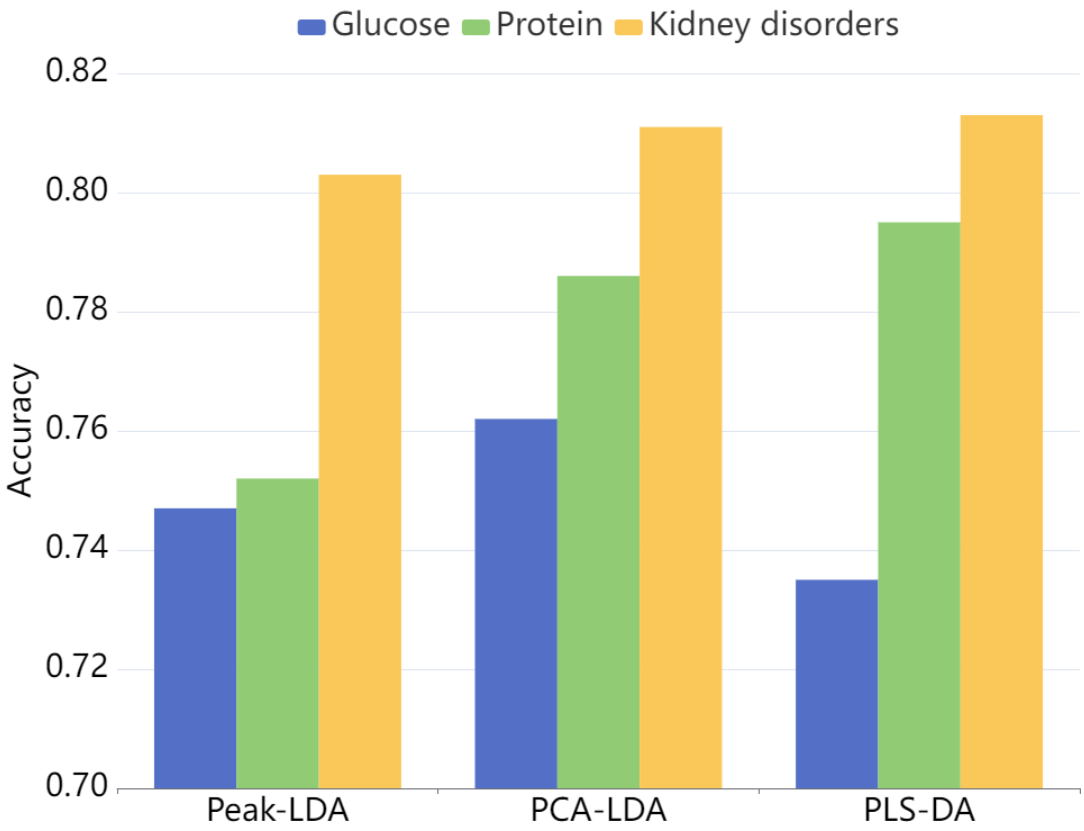


Figure S13: Accuracy results of Peak-LDA, PCA-LDA, and PLS-DA.

Note: We adopt the intensity of peak 937 cm^{-1} and 1558 cm^{-1} for linear discriminant analysis, named Peak-LDA, and compare the accuracy results with dimension reduction-based discriminant analysis, i.e., PCA-LDA and PLS-DA. Peak-LDA's performance is comparable to PCA-LDA and PLS-DA, demonstrating the validity of these two peaks.

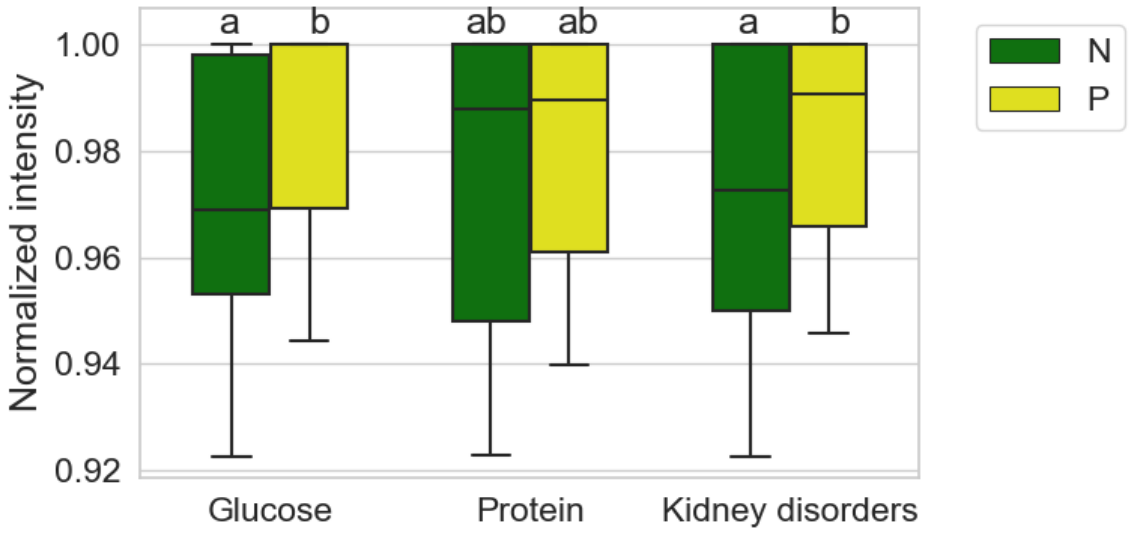


Figure S14: Normalized intensity of peak 937 cm^{-1}.

Note: Peak 937 cm^{-1} is the characteristic peak of glucose. Tukey honest significant difference (TukeyHSD) test is conducted for analysis of variance (ANOVA). The results show that there are significant differences ($p<0.05$) of glucose peak between glucose negative and glucose positive samples, also kidney disorders negative and kidney disorders positive samples. But there is non-significant difference ($p>0.05$) of glucose peak between protein negative and protein positive samples.

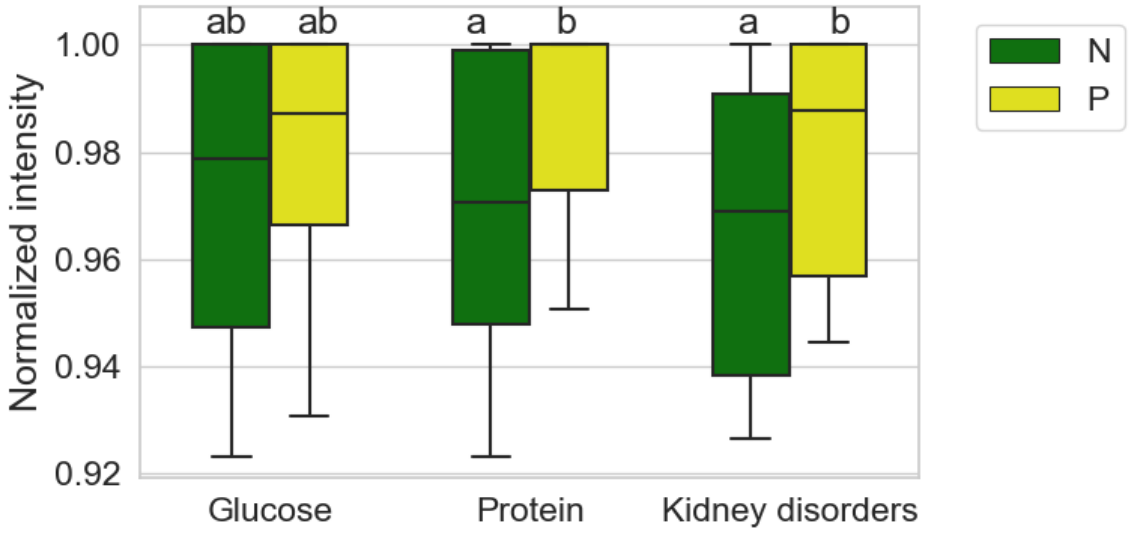


Figure S15: Normalized intensity of peak 1558 cm^{-1}.

Note: Peak 1558 cm^{-1} is the characteristic peak of protein. TukeyHSD test is conducted for ANOVA. The results show that there are significant differences ($p<0.05$) of protein peak between protein negative and protein positive samples, also kidney disorders negative and kidney disorders positive samples. But there is non-significant difference ($p>0.05$) of protein peak between glucose negative and glucose positive samples.

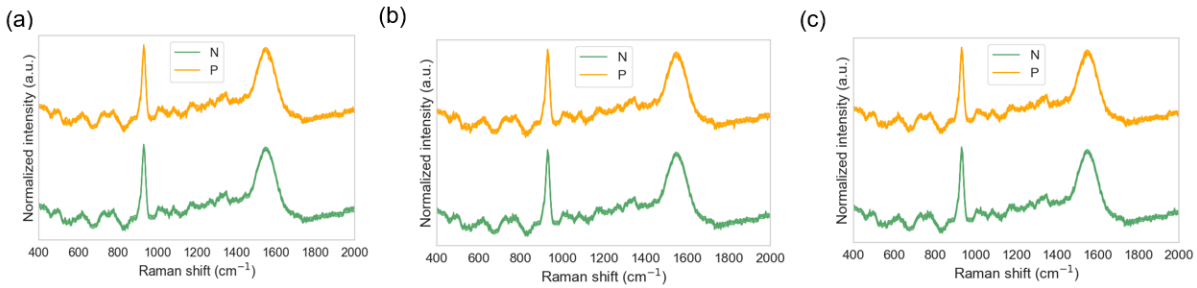


Figure S16: Five examples' spectra from the positive and negative samples of (a) glucose, (b) protein, and (c) kidney disorders.

References

[1] Chen, T.; Kornblith, S.; Norouzi, M.; Hinton, G. A simple framework for contrastive learning of visual representations. *Proc. Int. Conf. Mach. Learn.* 2020, *119*, 1597–1607.

[2] Lin, T.-Y.; Goyal, P.; Girshick, R.; He, K.; Dollar, P. Focal loss for dense object detection. *Proc. IEEE Int. Conf. Comput. Vis.* 2017, *16*, 2980–2988.

[3] He, K.; Fan, H.; Wu, Y.; Xie, S.; Girshick, R. Momentum contrast for unsupervised visual representation learning. *Proc. IEEE/CVF Conf. Comput. Vis. Patt. Recog.* 2020, *33*, 9729–9738.